

MORRIS, IL, USA

WMO: 720137

Lat: 41.425N

Lon: 88.419W

Elev: 584

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04867

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.9	(c) 1.4	(d) -10.8	(e) 3.1	(f) -3.0	(g) -5.7	(h) 4.1	(i) 3.1	(j) 29.8	(k) 31.8	(l) 26.8	(m) 25.3	(n) 6.4	(o) 290	(p) 0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.5	73.2	88.1	72.7	85.7	71.8	77.7	85.7	76.1	83.8	74.5	81.8	10.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	135.3	82.0	73.3	126.6	80.5	72.3	122.1	79.3	41.5	85.6	39.8	84.0	38.4	82.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.0	18.9	-10.3	94.5	6.7	2.2	-15.1	96.1	-19.0	97.3	-22.8	98.6	-27.7	100.2
			-11.4	80.9	6.6	1.3	-16.1	81.9	-20.0	82.7	-23.7	83.4	-28.4	84.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	24.3	26.2	39.2	50.0	61.8	70.7	73.1	71.1	66.0	53.1	40.4	29.2
	DBStd	19.75	12.05	12.01	10.80	9.13	9.15	6.01	5.30	5.04	7.44	9.48	9.43	10.82
	HDD50	2988	797	668	367	110	10	0	0	0	1	74	313	648
	HDD65	6117	1262	1086	801	454	173	16	4	9	75	387	740	1110
	CDD50	3194	1	2	33	111	377	622	716	654	481	172	23	2
	CDD65	848	0	0	2	5	75	188	255	198	106	19	0	0
	CDH74	7658	0	0	22	91	778	1688	2331	1590	966	192	0	0
	CDH80	2379	0	0	2	11	263	564	762	427	303	47	0	0
Wind	WSAvg	7.3	9.6	8.9	8.7	9.3	7.7	6.4	4.4	3.7	4.7	6.9	8.6	8.8
Precipitation	PrecAvg	36.5	2.0	1.6	2.3	3.7	4.4	4.5	4.2	3.9	3.1	2.8	2.6	2.0
	PrecMax	46.6	5.0	5.5	5.6	7.3	7.4	11.5	8.3	11.3	11.6	8.6	6.2	5.5
	PrecMin	23.9	0.4	0.4	0.5	1.4	1.0	1.0	0.3	0.6	0.7	0.4	0.3	0.4
	PrecStd	5.4	1.2	1.0	1.2	1.6	1.7	2.4	2.2	2.4	2.3	1.9	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	62.0	77.5	81.5	91.0	92.5	94.6	91.1	91.3	85.9	71.6	60.3
		MCWB	53.9	54.1	63.6	62.0	68.0	72.8	77.0	73.6	73.0	69.6	56.7	56.4
	2%	DB	49.8	53.8	67.3	76.7	86.7	89.9	89.9	87.8	87.9	79.4	63.8	53.4
		MCWB	47.4	48.5	58.0	59.8	67.4	72.5	76.1	73.9	72.5	63.3	55.3	49.8
	5%	DB	44.0	45.8	61.3	72.1	82.2	86.3	86.7	84.6	83.2	73.5	59.8	48.0
		MCWB	41.5	41.9	54.3	57.6	65.6	70.9	74.3	72.8	70.1	61.4	53.1	45.8
	10%	DB	39.2	40.9	56.1	67.5	78.2	82.4	84.2	82.0	80.1	69.2	54.7	43.4
		MCWB	37.0	37.9	50.0	55.9	64.5	69.0	72.9	71.4	67.2	60.8	49.0	40.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.9	56.3	64.1	65.8	73.1	78.8	80.8	78.7	77.2	71.9	61.7	57.7
		MCDB	55.8	59.1	73.2	76.7	82.4	87.5	88.4	86.2	86.2	81.7	66.9	60.1
	2%	WB	47.7	48.9	59.9	62.4	70.6	75.8	78.2	76.6	74.8	68.3	57.3	50.4
		MCDB	49.2	52.9	66.0	71.5	80.5	83.8	85.9	84.1	83.3	75.1	62.0	51.5
	5%	WB	41.7	42.8	56.0	59.9	68.7	73.8	76.6	75.0	72.3	65.3	53.6	45.7
		MCDB	44.3	46.0	60.7	69.1	78.3	81.9	84.2	81.8	79.4	70.5	59.1	47.1
	10%	WB	37.3	37.5	50.1	57.0	66.8	72.1	74.9	73.4	69.9	61.8	50.1	41.1
		MCDB	39.1	40.1	55.5	65.8	75.2	79.9	82.2	79.5	76.3	67.1	55.0	43.4
Mean Daily Temperature Range	5% DB	MDBR	14.2	15.2	17.0	21.1	21.2	20.0	19.5	19.5	21.9	20.3	16.7	13.4
		MCDBR	17.7	20.5	24.6	28.5	26.3	23.2	20.9	21.8	24.9	26.8	23.2	18.4
		MCWBR	15.1	15.9	16.0	15.5	11.8	10.4	10.1	10.7	11.1	13.5	15.6	15.7
	5% WB	MCDBR	17.3	20.1	22.0	25.7	21.9	18.9	18.8	18.2	20.3	20.8	20.1	17.1
		MCWBR	15.2	16.5	15.6	16.3	12.0	10.2	11.1	10.5	10.6	13.5	15.3	15.7
Clear-Sky Solar Irradiance	taub		0.292	0.311	0.346	0.400	0.430	0.437	0.453	0.442	0.412	0.355	0.311	0.292
	taud		2.430	2.381	2.381	2.240	2.211	2.234	2.181	2.203	2.258	2.419	2.492	2.451
	Ebn at Noon		274	285	286	276	269	267	261	260	260	265	265	264
	Edh at Noon		25	30	34	42	45	44	46	43	38	29	23	23
All-Sky Solar Radiation	RadAvg		526	809	1139	1479	1688	1920	1919	1693	1426	945	610	448
	RadStd		52	106	86	120	138	139	107	97	122	107	55	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page